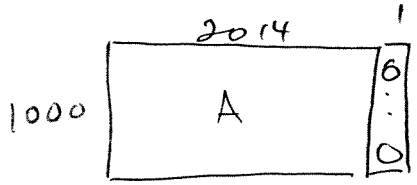
F14 Test 4 v3

1. If the coefficient matrix A in a homogeneous system of 1000 equations in 2014 unknowns is known to have rank 800, how many parameters are there in the general solution?

A. 200
B. 800
C. 1000
D. 1214
E. 2014
F. 0



parameters in gen'l soln = # variables − rank A

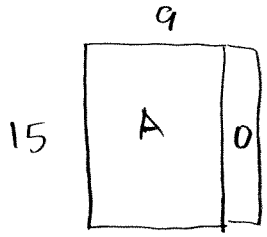
= 2014 − 800

= 1214.

2. Let A be the 15×9 coefficient matrix of a *homogeneous* linear system, and suppose that this system has infinitely many solutions with 6 parameters.

- What is the rank of A?
- Are the columns of A, considered as vectors in $\mathbf{R}^{15}$, linearly independent?

A. 0, Yes
B. 15, Yes
C. 15, No
D. 3, Yes
E. 3, No
F. 6, No



6 = # parameters in gen'l soln

= # variables − rank A

= 9 − rank A

∴ rank A = 9 − 6 = 3.

Write $A = [c_1 \cdots c_9]$, $x = \begin{bmatrix} x_1 \\ \vdots \\ x_9 \end{bmatrix}$. Then as $Ax = 0$ has only many soln,

$Ax = x_1c_1 + x_2c_2 + \cdots + x_9c_9 = 0$ does too. Hence

$\{c_1, \ldots, c_9\}$ is dependent

3. For a *nonhomogeneous* system of 3013 equations in 2014 unknowns, answer the following three questions:

I ○ Can the system be inconsistent? Yes
II ○ Can the system have infinitely many solutions? Yes
III ○ Can the system have a unique solution? Yes

A. Yes, Yes, No.
B. No, No, Yes.
C. Yes, No, Yes.
D. No, Yes, Yes.
E. Yes, Yes, Yes.
F. No, No, No.

I: e.g. $\left[\begin{array}{ccc|c} 0 & \cdots & 0 & 1 \\ \vdots & & \vdots & 0 \\ 0 & \cdots & 0 & 0 \end{array}\right]$ (3013 × 2014) is inconsistent

II: eg $\left[\begin{array}{ccc|c} 1 & 0 \cdots & 0 & 1 \\ 0 & & \vdots & 0 \\ \vdots & & & \vdots \\ 0 & \cdots & 0 & 0 \end{array}\right]$ (3013 × 2014) has only many solns (with 2013 parameters in the general solution)

III e.g. $\left[\begin{array}{cccc|c} 1 & 0 & \cdots & 0 & 1 \\ 0 & 1 & \ddots & 0 & 0 \\ 0 & 0 & 1 & 0 & \vdots \\ \vdots & \vdots & \ddots & 1 & \\ 0 & \cdots & \cdots & 0 & 0 \end{array}\right]$ (2014 × 2014)

If rank A = 2014 (as illustrated), the solution will be unique; the system indicated has the unique soln $(1, 0, \cdots, 0) \in \mathbb{R}^{2014}$.

4. Suppose $e, f \in \mathbf{R}$ and consider the linear system in x, y and z:

$$\begin{array}{rcrcrcr} x & - & y & + & ez & = & f \\ x & & & + & z & = & -1 \\ 3x & + & y & + & 2z & = & 1 \end{array}$$

a) If $[A\,|\,b]$ is the augmented matrix of the system above, find $\operatorname{rank} A$ and $\operatorname{rank}[A\,|\,b]$ for all values of e and f.

$$[A|b] = \left[\begin{array}{ccc|c} 1 & -1 & e & f \\ 1 & 0 & 1 & -1 \\ 3 & 1 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & -1 & e-1 & f+1 \\ 0 & 1 & -1 & 4 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & e-2 & f+5 \end{array}\right]$$

Hence $$\operatorname{rank} A = \begin{cases} 3 & \text{if } e \neq 2 \\ 2 & \text{if } e = 2 \end{cases}$$

and $$\operatorname{rank}[A|b] = \begin{cases} 3 & \text{if } e \neq 2 \text{ or } f \neq -5 \\ 2 & \text{if } e = 2 \text{ and } f = 5 \end{cases}$$

(Q.4 parts (b) and (c) are on the next page...)

4b). Using part (a), find all values of e and f so that this system has

(i) a unique solution, exists $\Leftrightarrow$

$\text{rank}\, A = \text{rank}\,[A|b] = 3.$

Hence a unique soln exists iff $e \neq 2.$

(ii) infinitely many solutions, or

There are infinitely many solutions iff

$\text{rank}\, A = \text{rank}\,[A|b] < 3$ iff

$e = 2$ and $f = -5$

(iii) no solutions.

The system is inconsistent iff $\text{rank}\, A < \text{rank}\,[A|b]$

hence iff $e = 2$ and $f \neq -5$

4c). In case b(ii) above, give a complete geometric description of the set of solutions.

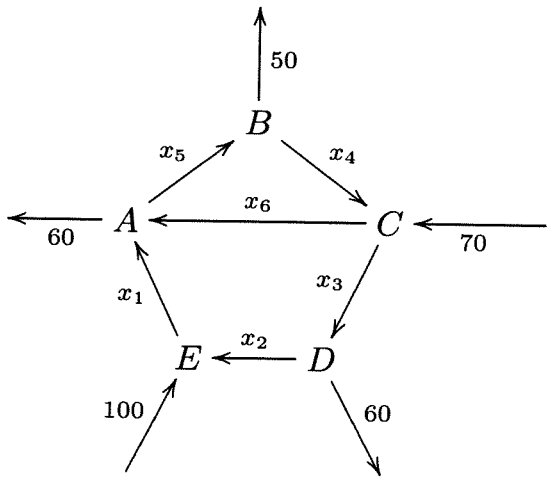
If $e = 2, f = 5,$

$$[A|b] \sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & +4 \\ 0 & 0 & 0 & 0 \end{array}\right]; \quad \begin{array}{l} x = -1 - \Delta \\ y = 4 + \Delta \\ z = \Delta \end{array}, \ \Delta \in \mathbb{R}.$$

Thus the general soln is a line in $\mathbb{R}^3$ through

$(-1, +4, 0)$ with direction $(-1, +1, 1)$.

5. Consider the network of streets with intersections A, B, C, D and E below. The arrows indicate the direction of traffic flow along the **one-way streets**, and the numbers refer to the **exact** number of cars observed to enter or leave A, B, C, D and E during one minute. Each x_i denotes the unknown number of cars which passed along the indicated streets during the same period.



(You must justify all your answers.)

a) Write down a system of linear equations which describes the the traffic flow, **together with all the constraints** on the variables x_i, $i = 1, \ldots, 6$.

(**Do not perform any operations on your equations: this is done for you in (b).** *Do not simply copy out the equations implicit in (b). You will not get any marks if you do this.*)

Flow in = Flow out

Intersection

A	$x_1 + x_6$	$=$	$60 + x_5$
B	x_5	$=$	$50 + x_4$
C	$x_4 + 70$	$=$	$x_3 + x_6$
D	x_3	$=$	$x_2 + 60$
E	$x_2 + 100$	$=$	x_1

One way streets $\Leftrightarrow$ $x_i \geq 0$, $i = 1, \ldots, 6$

"# cars" $\Leftrightarrow$ $x_i \in \mathbb{Z}$.

(Q.7 part (b) is on the next page.)

5(b). The reduced row-echelon form of the augmented matrix of the system in part (a) is

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & -1 & 1 & 60 \\ 0 & 1 & 0 & 0 & -1 & 1 & -40 \\ 0 & 0 & 1 & 0 & -1 & 1 & 20 \\ 0 & 0 & 0 & 1 & -1 & 0 & -50 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Give the general solution. (Ignore the constraints from (a) at this point.)

$x_1 = 60 + \Delta - t$
$x_2 = -40 + \Delta - t$
$x_3 = 20 + \Delta - t$
$x_4 = -50 + \Delta$
$x_5 = \Delta$
$x_6 = t$

; $\Delta, t \in \mathbb{R}$

5(c). If $\overline{ED}$ were closed due to roadwork, find the minimum flow along $\overline{AC}$, **using your results from (b).**

If $\overline{ED}$ is closed, $x_2 = 0$, so $\Delta - t = 40$ or $t = \Delta - 40$.
The flow along $\overline{AC}$ is $x_6 = t$, which attains its minimum when Δ is a minimum. We examine the constraints:

$x_1 \geq 0$, but $x_1 = 60 + \Delta - t = 100$ ✓

$x_2 = 0$ ✓

$x_3 \geq 0$, but $x_3 = 20 + \Delta - t = 60$ ✓

$x_4 \geq 0 \iff \Delta \geq 50$
$x_5 \geq 0 \iff \Delta \geq 0$
$x_6 \geq 0 \iff t \geq 0$

These show that the minimum value of Δ is 50, so the minimum value of $x_6 = t = 10$.

Hence the minimum flow along $\overline{AC}$ is 10.

6. State whether each of the following statements is (always) true, or is (possibly) false, in the box after the statement.

- If you say the statement may be false, you must give an explicit example - with numbers!
- If you say the statement is always true, you must give a clear explanation.

a) A linear system of 5 equations in 4 unknowns is always inconsistent.

$A = 0,\ b = 0 : 5$ $\left[\begin{array}{c|c} 0 & 0 \end{array}\right]$ (4 columns, 1 column) has infinitely many solutions (with 4 parameters in the general solution)

or $\left[\begin{array}{c|c} 1\ 0 \cdots 0 & 0 \\ 0 \quad\ \ \vdots & \vdots \\ 0 \cdots 0 & 0 \end{array}\right]$ has infinitely many sol'n (with 4 parameters in the gen'l sol'n)

ANSWER **FALSE**

b) If the coefficient matrix of a homogeneous system has a column of zeros, the system has infinitely many solutions.

If A has a column of zeros, then rank $A <$ # cols of A, so the system $Ax = 0$ has infinitely many solutions (with at least one parameter in the general solution.)

ANSWER **TRUE**

c) If the RRE form of the coefficient matrix of a homogeneous system has a row of zeros, the system has infinitely many solutions.

e.g. $\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$ has a unique soln (namely, $(0,0)$).

ANSWER FALSE

d) If A and B are 2 by 2 matrices then $AB = BA$.

Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

then $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, while

$BA = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \neq AB$

ANSWER FALSE

8. [Bonus] If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 matrix such that the vectors $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$ are linearly independent, prove carefully that rank $A = 2$. (You cannot choose the matrix A - your proof must work for every 2×2 matrix with the property above, i.e. every 2×2 matrix with independent columns.)

Write $A = [c_1 \; c_2]$ with $c_1 = \begin{bmatrix} a \\ c \end{bmatrix}, c_2 = \begin{bmatrix} b \\ d \end{bmatrix}$

Since $\{c_1, c_2\}$ is l.i., then $x_1 c_1 + x_2 c_2 = 0$

$\Rightarrow x_1 = x_2 = 0$. Thus if $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$,

$$Ax = x_1 c_1 + x_2 c_2 = 0 \Rightarrow x = 0$$

Thus rank $A = 2 = \#$ cols of A, since homogeneous linear system $Ax = 0$ has a unique sol^n.

7. ~~8.~~ [Bonus] If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 matrix such that the vectors $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$ are linearly independent, prove carefully that rank $A = 2$. (You cannot choose the matrix A - your proof must work for every 2×2 matrix with the property above, i.e. every 2×2 matrix with independent columns.)

Since $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$ are l.i., neither is $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

If $a \neq 0$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \sim \begin{bmatrix} 1 & b/a \\ 0 & d - bc/a \end{bmatrix}$.

If $d = \frac{bc}{a}$, then $\begin{bmatrix} b \\ d \end{bmatrix} = b/a \begin{bmatrix} a \\ c \end{bmatrix}$, contradicting the linear independence of the columns. Hence, if $a \neq 0$, $A \sim \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix}$ so rank $A = 2$.

If $a = 0$, then $c \neq 0$. So $A = \begin{bmatrix} 0 & b \\ c & d \end{bmatrix} \sim \begin{bmatrix} 1 & d/c \\ 0 & b \end{bmatrix}$. If $b = 0$, then $\begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ d \end{bmatrix} = d/c \begin{bmatrix} 0 \\ c \end{bmatrix} = d/c \begin{bmatrix} a \\ c \end{bmatrix}$, contradicting the l.i. of the columns. Hence $b \neq 0$ and so $A \sim \begin{bmatrix} 1 & d/c \\ 0 & 1 \end{bmatrix}$ has rank 2.

Hence, in all cases rank $A = 2$.